Supplementary Evidence: When Do LLM Side-Channel Defenses Protect Agent Workflows?

This supplement provides the search accounting, corpus maps, saved empirical results and source-bound audits accompanying the main paper. The package README maps these sections to machine-readable data and the fit-free verifier. Original failed and negative outcomes are preserved alongside positive results.

S1. Search and Evidence Audit

TABLE S1. HISTORICAL ROUTE ACCOUNTING FOR THE ORIGINAL OCTOBER 8, 2026 SNAPSHOT, INCLUDING ITS LATER SOURCE CHECKS. COUNTS OVERLAP AND DO NOT FORM A COMPLETE PRISMA FLOW; THE OCTOBER 21 SUPPLEMENTARY ROUND IS REPORTED SEPARATELY.

Route / disposition	Count
Broad IEEE Q4 hits; not fully screened	35,743
Amended Q4b export; recall not verified	847
IEEE saved export occurrences / distinct records	2,079 / 2,001
IEEE full-text referrals / background routes	25 / 205
ACM saved occurrences / publication groups	16,687 / 13,201
ACM Oct. 14 provisional exclusions / full-text referrals / unresolved	9,368 / 883 / 2,950
ACM dispute groups / combined check candidates (Oct. 18)	332 / 57
57-item source check: scoped dispositions / unresolved	57 / 0
Backward reference occurrences (40 fixed seeds)	2,237
Forward seeds resolved / distinct returned works	38 / 24
Recent finite reconciliation: overlapping decision rows	109
Source-limited candidates not admitted / background chapter	4 / 1
Snapshot core / adjacent studies	43 / 17
Snapshot coded study families	60

The 57 scoped dispositions are not completed full readings. B069's complete official abstract supports background routing; later-page eligibility remains unchecked. The 109 rows overlap; 883/2,950 are dated global routes. Q4b does not replace unscreened Q4. This snapshot's coded-set cutoff is October 8, 2026; its counts do not describe the later expanded collection.

Targeted discovery beyond native exports. The saved September 12 query families include `"agent" "traffic" "fingerprinting"` with an arXiv domain filter, `"MCP" "timing" "side channel"` with the same filter, and `"LLM" "side-channel" "2026"` with USENIX/NDSS filters. October 8 follow-up includes `"agent" "side channel" "2026" privacy workflow` and exact-title checks for AgentPrint, UI Traces and FP-Agent. Primary PMLR records were also checked for prompt-cache evidence. These web-discovery routes yielded source leads, not enumerated native hit exports or a venue-completeness denominator. The companion search logs distinguish discovery, primary-source inspection and subsequent family decisions; no later source lookup is relabeled as native query recall.

Acquisition and grouping. Table S2 prints the actual native queries. Export grouping uses normalized DOI, then normalized title only without a conflicting explicit DOI; study-family/version merging is a later source decision. The acquired IEEE frame has 2,079 occurrences/2,001 distinct records, and ACM has 16,687 occurrences/13,201 groups. ACM final-named exports were byte/hash matched; the Q4b June and Q5 count discrepancies remain unresolved. Historical broad Q4 membership cannot be reconstructed from its later first-1,000 prefix.

Finite accounting versus coverage. Every prepared metadata ID has a disposition: 4,753 unique IDs comprise 2,778 exclusions, 116 referrals and 1,859 unresolved routes. Original answer artifacts include 4,653 schema-valid IDs, 80 format holds and 20 missing finals. Later source reconciliation of 1,775 IDs distinguishes 1,313 containers, 455 scoped dispositions and seven material holds; it is not 455 full readings. The 109-row nominated source map overlaps these and citation routes. The complete row-level decisions and source scopes are in the companion package; none of these frames is a subtractive full-search PRISMA flow.

Known papers and exclusion checks. An exact-identity replay against the saved 2,001-record IEEE and 13,201-group ACM frames finds 1/24 and 0/24 fixed known papers. Online indexing and native field eligibility remain unverified. Literal title/abstract term matching does not emulate native stemming or indexed fields: even the retrieved Early Bird record fails that strict literal test for its saved branches. We therefore report export presence, not recall or evidence that nonhits are ineligible.

The earlier frozen 1,555-record exclusion frame has 1,300 checks (1,075 keyword-risk and 225 stratified): 1,297 retained abstract exclusions, two referrals and one missing abstract. One referral was later excluded against a preprint; two material checks and 255 unsampled lower-risk records remain. Separate later checks cover 33/554, 16/144 and 20/2,090 routes, yielding 3, 2 and 5 source/material holds.

TABLE S2. EXECUTED NATIVE QUERIES AND SAVED EXPORT OCCURRENCES. IEEE DATES: OCTOBER 8, 2026; ACM QUERY/PARTITION ACQUISITION: OCTOBER 8–9, 2026. COUNTS ARE NOT INCLUDED STUDY FAMILIES.

ID	Exact query	IEEE saved/hits	ACM saved/original hits
Q1	`("large language model" OR LLM) AND ("side channel" OR "side-channel" OR "timing attack")`	63/63	479/479
Q2	`(agent OR agentic) AND ("traffic analysis" OR fingerprinting OR "metadata leakage" OR "side channel" OR "side-channel")`	671/671	6,181/6,181
Q3	`("KV cache" OR "prompt caching" OR "speculative decoding") AND (privacy OR leakage OR attack)`	37/37	511/511
Q4	`("browser agent" OR "research agent" OR "tool calling" OR "model context protocol" OR "multi-agent") AND (timing OR fingerprinting OR leakage)`	0/35,743	Not run
Q4b	`(("browser agent" OR "research agent" OR "tool calling" OR "model context protocol" OR "multi-agent") AND (timing OR fingerprinting OR leakage) AND ("large language model" OR LLM))`	847/847	4,026/4,026
Q5	`("language model" OR LLM) AND ("traffic shaping" OR "cache isolation" OR padding OR "side-channel defense")`	95/95	3,689/3,688
Q6	`("language model" OR chatbot OR "AI assistant" OR "generative AI") AND ("side channel" OR "side-channel" OR "traffic analysis" OR fingerprinting OR "timing leakage")`	366/366	1,801/1,801

IEEE used Command Search/default metadata, without a lower-year or document-type filter; every result page was exported as RIS with Citation and Abstract. Q4's zero means not exported in the October 8 acquisition, not zero hits or a completed screen. Q4b supplements Q4; recall preservation is unverified. The later Q4 prefix does not reconstruct historical membership. ACM used the All field in the ACM Full-Text Collection (not Guide), then native E-Publication Date partitions and All Results/BibTeX export. Partition receipts retain exact dates and URLs. Q4b's June bin has 285 displayed versus 284 saved; agreement of the overall saved/original counts does not repair the missing identity. Q5's partition total differs from its original whole-query count. Final-named export files were byte/hash matched on October 9; these two retrieval discrepancies remain. The October 8 coded-set cutoff is checked separately from acquisition partitions.

The last covers only 10/1,074 keyword-risk records. These AI-assisted checks are not interchangeable frames and do not estimate false negatives. The 23-case blinded AI check has 17/21 determinate agreement; the 332-dispute readbacks yield 57 source-scoped dispositions, not human double coding.

Final source dispositions and counterconditions. Four source-limited candidates are not admitted to the present core analysis: Wiretapping LLMs (primary full text not obtained); Entropy Based Obfuscation (publisher PDF saved, but adapted/simulated tables lack established new-result lineage); the memory-bandwidth poster (empirical report not obtained); browser-panel analysis (auxiliary versus direct disclosure not established from accessible primary material). B069 is background on its complete official abstract; later pages were not obtained. These are final dispositions for this search, not assertions that unavailable evidence does not exist.

The source checks seek counterconditions to broad deployment claims: AHB's non-Fake online point prevents extrapolating one combined-condition task loss; UI's matched delayed-view adaptation preserves a positive source-local judgment; SpliceDefense and the cache prototypes preserve applicable component evidence. Unacquired broad results and incompletely audited exclusions limit coverage. The query/partition receipts, source dispositions and exact known-paper checks preserve that boundary without treating another search as a prerequisite to reporting the supported comparisons.

S2. Family Index, Corpus Maps and Prior Work

The accompanying *Supplementary Evidence* archive is the access route for the full tables and fit-free checks. Its README starts with the current paper, family index, result ledger, utility/cost inventory and claim links. Each result joins a citation-key family, source version/hash and locator to its receiver/intervention fields and supported judgment; Table S3 is a filled example. Separate case-evidence entries identify additional source subconditions, rather than silently pooling them into the 33 scoped inventory records.

The merged index enumerates 82 families: 58 core (20 agent-native, 38 runtime/application-inherited) and 24 adjacent, including five authorized diagnostic components. The original 60-family index is a separately labeled historical snapshot. Mixed semantic attribution remains adjacent under the auxiliary-only rule. The package preserves the snapshot's 33 scoped result records/66 field positions and the complete original empirical and workflow analyses. S10 supplies the 22 added-family rows and expansion sensitivity. Source locators and hashes support inspection; third-party full texts/raw traces are obtained from their named sources.

Roles of the remaining indexed sources. Additional core studies concern content/hidden-property inference [1], [2], [3], [4], [5], [6], identity/provenance attribution [7], [8], [9], [10], and authorized integrity diagnostics [11], [12], [13]. Adjacent audit, execution, browser-data and profiling studies supply boundary comparisons rather than additional auxiliary privacy attacks [14], [15], [16], [17], [18], [19], [20], [21]. The index records each receiver, target relation and evidence scope; the reference list includes these indexed

TABLE S3. FILLED RECORD AND ITS LICENSED JUDGMENT. VALUES ARE SOURCE-REPORTED; NO CROSS-MODE EFFECT IS COMPUTED.

Join key / field	Entry
Family / result	`dangol2026workflow` / `DANGOL-livebinding`.
Version / locator	arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [22].
Target / view	Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation.
Intervention	Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures.
Workload / testbed	Workflow classes with client, relay, gateway and agent co-located on one loopback host.
Outcome / cost	Accuracy 0.30; about 13× bytes and 11× latency in that condition.
Unresolved fields	Objective terminal task utility NR-R; precise paired live-refit comparison NR-V.
Licensed judgment	These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter.

sources, not just the central case papers.

S2.1. RQ sensitivity

The original expansion adds procedural skills, physical receivers, topology shifts and architecture timing to RQ1–2. The latest finite search additionally changes the positive defense evidence: CachePrune and OTRO supply useful mechanism-bound privacy/quality/performance conditions. RQ3 still requires generator-specific bridges; RQ4 separates simulation, component overhead, fixed guardrails and executed protection. S10 reports these changes explicitly. Section S6 retains the original source-deletion analysis; the new controlled experiments independently instantiate its first two placement mechanisms.

S3. Retained Failure-Inclusive Controls

The 288-trace fixed-policy pilot has call-count-determined labels and saturated coarse F1=1.00, so it supports no sequence-increment claim. In the separate tool-front-end study, the observer sees both tools; the constrained baseline succeeds on 19/24 tasks, below the frozen promotion gate. These local results are not passive-network or deployed-defense evidence.

The retail case retains 148 windows, 49 empty observations and 102 stopped-error episodes. All three frozen loss contrasts have intervals including zero; test goal achievement is 7/40 versus 8/40 and joint utility remains unresolved. Neither this null-compatible estimate nor failed tasks establish absence of leakage or effective defense. Full protocols and results remain in the delivered evidence package.

S4. Public-Trace Reanalysis: Protocol and Boundaries

UI Traces (commit `4f3df18`) supplies 4,200 records/14 declared identities, retaining 1,806 error/absent-result indicators. Labels are declarations, not authenticated backends or task correctness. The supplemental package binds source bytes, code and receipts.

Predictors use JS types, local time and callback order, excluding collector time/navigation, URL/DOM text, questions, answers and diagnostics. Cross-document order assumes an archive receiver, not authenticated website capture. Exact-question-disjoint train/validation/test groups have 2,043/1,022/1,135 rows and 395/202/226 questions (823 distinct questions). The source split identifies FRAMES but has no semantic task-family field. Full identity–dataset/exact-question support and a seven-bin first-word *grammatical proxy*, explicitly not a semantic family, are supplemental; neither semantic independence nor family balance is established.

The original 19-fit comparison selects weighted logistic/forest candidates on validation loss only; selected models are forests. H2 adds type bigrams to coarse counts/page-age. Original intervals use 3,000 paired-question resamples and a nominal 98.333% three-contrast allocation, conditional on fixed fitting/selection and noise realizations. Archived H1's identical selected predictor yields exactly zero difference and interval $[0, 0]$; it is a selection illustration, not equivalence evidence.

For Q questions with n_q records each, loss is $Q^{-1}\sum_q n_q^{-1}\sum_{i\in q} -\log p_{i,y_i}$; accuracy substitutes the correct-argmax indicator. Fit weights $N/(Qn_q)$ equalize question weight. Add $\epsilon = 10^{-6}$ to every class probability, renormalize, then clip true-label probability to $[\epsilon, 1]$. Bootstrap draws Q questions with replacement; original contrasts share draw indices, with percentile endpoints 0.8333% and 99.1667%.

Controls declared with original outcomes known. The additive protocol is frozen before new fits, with original outcomes known. All-14 common support is empty. Select the largest identity subset with at least 20 common training questions, then greatest support and lexical tie-break: Gemini 3.1, Gemma 4-31B, Seed 2-lite, with 32/10/10 questions. Hash order selects one row per question/identity, retaining failures; test has 30 rows. This is a three-class sensitivity result.

The first-Wikipedia-document *proxy* stops at HTTP navigation, beforeunload, unmarked origin/URL change or clock reset, but not SPA history alone. Markers gate membership and never enter features. No frame/document IDs authenticate membership; late callbacks may cross boundaries. Empty views remain and epochs are not pooled. The supplement gives the complete six-event projection example and boundary implementation.

Three row-seeded shuffles preserve counts/timestamps/coarse features in *all partitions*, refitting combined candidates with coarse retained. Fits total 8 matched-task, 8 document-proxy, 8 clean and 12 shuffled;

TABLE S4. PRIMARY OBSERVABLE/RECEIVER PROFILE × SIGNAL ORIGIN IN THE 43 CORE FAMILIES OF THE ORIGINAL OCTOBER 8, 2026 SNAPSHOT. COLUMN n COUNTS DISTINCT FAMILIES; EACH FAMILY OCCUPIES ONE PROFILE ROW BUT MAY HAVE MULTIPLE ORIGINS. ORIGIN CELLS COUNT FAMILIES, NOT RESULTS, AND THEREFORE NEED NOT SUM TO n. EXTRA OBSERVER VIEWS REMAIN IN THE SNAPSHOT LEDGER. LATER ADDITIONS ARE OUTSIDE THIS MAP; ZEROS ARE NOT FIELD-WIDE GAPS.

Observable / receiving permission	n	P1	P2	P4	P5	P6	P7	P8	Representative source
Flow/destination	12	0	8	0	5	1	2	2	AgentPrint; WRA; NetEcho
Own-site actions/attributes	5	0	0	0	5	0	5	0	UI Traces; FP-Agent
API duration/cache/oracle	12	0	12	0	0	0	0	0	Early Bird; LeakyLMs
Resource/physical	10	10	3	0	1	0	0	0	TDXRay; ClawGuard; LLM RCA (diagnostics)
Trajectory/procedure/log	4	0	0	1	4	2	0	0	SIGLEAK; cycle detection (diagnostic)
Total distinct families	43	Each family contributes once to n.							

P1 hardware; P2 serving; P4 application/API; P5 orchestration; P6 tools; P7 environment; P8 multi-agent. P3 denotes transport; this collection codes it as a receiving layer but has no P3 generating-origin entry. This does not rule out transport-generated signals elsewhere. Composite origins are multi-label descriptions, not independent channels. Privileged logs and physical diagnostic receivers are not ordinary website capabilities.

TABLE S5. FROM PRIOR CROSS-LAYER SYSTEMATIZATIONS TO RESULT-LEVEL DEPLOYMENT DECISIONS: THE SPECIFIC COMPARISON OBJECT AND ITS DEMONSTRATED USE.

Prior work	Established perspective	Added result-level question	Demonstrating analysis
Banerjee et al. [23]	Cross-layer compound AI attacks can relax threat assumptions; MITRE organization	Does cover end at the protected interface, get reserialized, or execute as an action?	NetEcho final-hop check and Dangol/AHB transport/action distinction; explicit conditions for permitting transfer.
Forough et al. [24]	Confidential-computing threat model; partial inference-defense transfer; isolation is not side-channel immunity	Which oracle, execution mode and component-cost result licenses the proposed defense transfer?	Early Bird KV/semantic split and KVGov hardware/simulation/analytic separation.
Cai et al. [25]	Traffic features, shaping objectives and bandwidth trade-offs	Can a traffic-cover cost/utility argument be used for a dummy tool action?	Dangol/AHB: a state-machine boundary determines which task/policy outcomes must additionally be measured.
Debenedetti et al. [26]	Auxiliary ML components induce privacy channels	Distinguish an execution-mediated oracle from ordinary semantic attribution	C3 boundary and mixed-output eligibility judgments.
Lahjouji–Colaco [27]	Agent data surfaces, governance, composition and cross-session risks	Which bridge test would license a specific transfer?	Receiver/intervention and generator/holdout result maps.
Kim; Horváth et al. [28], [29]	Broad agent security; physical DNN mechanisms/threat models	Compare source and receiving privileges without a universal strength order	Remote calibration versus own-site, local and privileged diagnostic profiles.

all 36 finish before test prediction, without convergence warnings. Clean H2 reproduces exactly. Each shuffle selects coarse: zero increment is then an algebraic consequence of selecting the same predictor. This checks the selection pipeline, not whether shuffled sequence features contain no information; a forced-sequence comparison would be a different, unperformed analysis.

New intervals use 3,000 question resamples, seed 20261018 and 98.333% allocation across the three new contrasts, conditional on fits and shuffle streams. They are exploratory, not blind confirmation or global adjustment across all historical analyses. Separate code reconstructs points/intervals from saved probabilities. Matched-task uncertainty and the 509/1,135 test error flags (3.4–91.5% by identity) prevent intrinsic-identity claims.

The saved-result verifier supplies local replay; it does not establish independent replication or live utility/cost.

Failure/activity diagnostics. The configuration was frozen with prior outcomes known: three representations, four candidates each, original weights, seed 20261008; all 12 fits precede test prediction. Failure-only uses error/absent-result bits; empty-only uses no projected events; activity-only uses log-count plus seven page-age summaries, not cross-document duration. Candidate hyperparameters and projections are supplemental. No new confirmatory intervals are supplied. Flags cover 43.00% overall and 44.85% test rows, not authenticated task success. Stratifying 509 flagged/203 questions versus 626 unflagged/216 questions changes support; increments are not a causal failure effect. Semantic-family balance remains unavailable.

Exact H3 perturbation and adaptation. The unchanged perturbation sets $t'_{ire} = t_{ie} + u_{ire}$, $u_{ire} \sim U(0, 5000\,\mathrm{ms})$, independently per valid page-local timestamp in source-array order. Missing times consume no draw; types/order/page keys remain fixed. Three row-seeded realizations transform all partitions. Each adaptation clones the clean-selected representation/architecture and fits once on perturbed training rows with identical weights, without retuning or additional data (three fits). Both programs predict the same perturbed test rows; average paired losses across realizations, rows within question, then questions. The supplemental exact protocol supplies entry points, seed

TABLE S6. SOURCE–RECEIVER–INTERVENTION MAP. THE TARGET COLUMN DISTINGUISHES CONTENT/PROPERTY INFERENCE, ATTRIBUTION AND HUMAN–AGENT DISTINCTION; NONE ALONE ESTABLISHES AN UNAUTHORIZED PRIVACY VIOLATION. U DENOTES AN UNRESOLVED SOURCE/RUN BINDING.

Source / target	Signal origin → receiver	Intervention / evaluated mode	Surviving path or restriction	Comparison consequence
UI Traces / identity attribution	Agent actions → own-site JS	Recorded timestamp delays; offline D2	Types and order preserved	Test residual action features; no live task-cost estimate.
NetEcho / content inference	Serving → network A/B traces	Upstream padding/batching; protected-data fit	Decoded upstream content/state can be reserialized downstream	Inspect final-hop emission; D1 is not automatically D2.
SIGLEAK / skill nondisclosure	Procedure → task/output receiver	Common prompt defense	Procedural signatures remain	Compare same receiver/target; not a paired installation effect.
AHB / human–agent distinction	Touch behavior → app/detector	Executed fake touches; separate transforms	Task state may change; fit binding U	Measure transitions and objective utility; do not import offline delay costs.
WRA / private-intent inference	Real/decoy research requests → visited domains	Concurrent execution; interleaved-domain evaluation	Intent reconstruction falls with one decoy	Preserve domain-view privacy benefit; terminal decoy utility and cost are separately unreported.
Dangol / workflow-property inference	Protocol/traffic → defined link	Stored captures versus live OHTTP (§9.7, OHTTP paragraph after Table 5)	Feature ablation differs from network masking	Keep accuracy 0.17 capture and 0.30 live populations separate.
Early Bird / cache state	Cache reuse → API timing	KV simulation versus GPT-Cache prototype	Distinct matching mechanisms	6ms/4% applies to semantic prototype only.
KVGov / private prompt	Shared cache → probe oracle	Simulated inference; real timing; analytic isolation	Hash-key isolation not semantic-index isolation	Hardware gap does not become deployed defended ASR.
FP-Agent / identity attribution	Policy/browser → visited site	Task/site evaluation, mixed attributes	Static configuration may remain	Task holdout is not independent-visitor holdout.
Traffic benchmark / topology	Multi-agent workflow → selected links	Explicit topology/view shift	Unseen generators differ from hidden links	Cross topology with receiving-view restriction.
LeakyLMs / architecture	Inference → streaming client	Chosen-prompt timing; no measured defense	Same-GPU-class calibration for architecture	API reachability is not hardware-independent recovery.

TABLE S7. DEPENDENCY–HOLDOUT–CLAIM MAP. EACH ROW EXCLUDES ONE EXPLANATION, NOT ALL CONFOUNDING.

Dependency / source	Required holdout	Claim licensed; residual limit
Task / UI Traces	Question/task-family	New task units; fixed harness remains.
Browser / FP-Agent	Visitor/configuration	Configuration transfer; task-only split does not test it.
Workflow / traffic benchmark	Agent topology or composition	Tested topology shift; no all-orchestrator guarantee.
Hardware / LeakyLMs	GPU/serving calibration	Tested architectural transfer given calibration; not unknown-hardware recovery.
Repository / coding fingerprints	Repository and generation pathway	Tested attribution domain; not covert human-mediated use.
Order / worked case	Same units, refit shuffle	Fixed-program order increment; not task-independent identity.

TABLE S8. FIXED-ARCHIVE PREDICTIVE EVIDENCE AND MATCHED SUPPORT. UNITS: NATS OF QUESTION-WEIGHTED LOSS REDUCTION. POSITIVE VALUES FAVOR THE SECOND PROGRAM. INTERVALS CONDITION ON FITTED PROGRAMS; THE ORIGINAL AND ADDITIVE-CONTROL ALLOCATIONS REMAIN SEPARATE.

Contrast	Estimate	98.333% interval
Worked case: coarse vs. combined (H2)	0.3447	[0.2974, 0.3944]
Matched questions, 3 identities[a]	0.2634	[−0.2584, 0.7241]
Reanalysis: clean vs. adapted (H3)[b]	0.1503	[0.1205, 0.1815]

[a] 30 rows/10 test questions, 3 classes, rather than 1,135 rows/226 questions, 14 classes; not a direct effect-size comparison. Its interval uses the additive three-control allocation. Archived identical-predictor H1 remains in the protocol.
[b] Both programs predict the same timestamp-perturbed test rows; three fixed realizations are averaged.
Cross-document order assumes an archive receiver, not verified ordinary-website capture. Intervals vary test-question weights, not training, task families or deployments. No live defense utility or cost is inferred.

strings and invariants. The transformation edits stored times; it does not execute action delays.

S5. Intervention Inventory and Evidence Dispositions

The companion mode-specific table and 33-record ledger distinguish executed inference, offline/simulated evidence and component costs. All 66 utility/cost positions retain the

TABLE S9. PUBLIC-TRACE PERMISSION PROJECTION. "SITE" MEANS ITS OWN DOCUMENT ONLY; IT DOES NOT AUTHENTICATE CROSS-DOCUMENT CAPTURE. FULL FIELD-LEVEL CSV AND ONE COMPLETE INGESTED-RECORD PROJECTION ARE SUPPLEMENTAL.

Field	Site JS	Analysis use
Type, page-local time	Yes	Original predictors; activity diagnostic
Within-document order	Yes	Type bigrams
Cross-document order	Unverified	Archive-receiver bigrams
URL/origin	Own page	Origin gate only; no text/page-key feature
HTTP navigation, collector time	No	Boundary proxy only / time excluded
Error/result metadata	No	Analyst-only failure baseline
Task/answer metadata	No	Excluded; question hash splits/weights only
DOM/link text	Own page	Excluded from all predictors
Document/frame ID	Absent	No authenticated document membership

TABLE S10. POST-REVIEW H2 CONTROLS. POSITIVE VALUES DENOTE A QUESTION-MEAN LOG-LOSS REDUCTION (NATS); BRACKETS ARE CONDITIONAL 98.333% RESAMPLING INTERVALS. NOT A NEW AGENT RUN.

Contrast	Test rows / questions	Effect [interval]
Matched tasks, 3 identities: coarse minus sequence	30/10	0.2634 [−0.2584,0.7241]
First-document-epoch *proxy*: coarse minus sequence	1135/226	0.1504 [0.1193,0.1825]
Original clean minus mean of 3 refit shuffle increments	1135/226	0.3447 [0.2961,0.3953]

All 14-class common support is empty. The first-document view is reconstructed, not authenticated. Its 0.1504 feature increment and H3's 0.1503 timestamp-adaptation effect are distinct contrasts on the same archive, not independent replications. Cross-document order assumes an archive receiver. All three shuffles select coarse, so their zero increments make this contrast's 0.3447 point equal to original H2. The intervals use separate frozen bootstrap seeds (20261018 here; 20261008 for original H2), yielding different finite-resampling endpoints. Zero selected increments do not establish information equivalence. No task utility or live defense is measured.

source, scope, state and consequence for a named claim or inventory-only role. The main deployment-question table does not treat these heterogeneous records as a common success/failure denominator.

Four textual defense assertions lack the quantitative binding needed for a matched effect (Table S12). Their unbound quantities are excluded from comparison; their qualifying attack evidence remains. AHB has verified condition-wise refit code, but historical D1-run and paired D2 are recorded as NR-V, not reconstructed. Its task outcome is not inferred from that code.

D1 denotes protected-condition fitting/calibration; D2 denotes a paired training-procedure comparison on the same defended test condition. Neither entails feasible labeled-data access or matched budgets. Token payload, plaintext padding, wire bytes, model loading, map lookup and added service latency remain distinct cost objects. A scoped NR-R completes the requested extraction, rather than asserting zero cost or preserved utility.

TABLE S11. ADDITIVE FAILURE/ACTIVITY DIAGNOSTICS ON ALL 1,135 TEST ROWS/226 QUESTIONS. POINT SCORES USE THE ORIGINAL QUESTION WEIGHTS; NO NEW CONFIRMATORY INTERVALS. ORIGINAL PROGRAMS ARE READ FROM SAVED PROBABILITIES, NOT REFITTED.

Program	Loss (nats)	Accuracy (%)
Training prior (saved)	2.64267	6.59
Error/result flags only[a]	2.47464	14.25
Projected-empty flag only	2.63832	6.79
Activity/page-age only[b]	1.81732	37.68
Original coarse (saved)	1.44058	51.29
Original coarse+sequence (saved)	1.09591	66.26

[a] Analyst-only diagnostic, not an observer-valid attacker; two binary flags, not error text. [b] Total event count and seven local-page-time summaries; no action types/order, diagnostics, text or session duration. Same four candidate architectures and validation-only selection for each of the three new representations (12 fits). Low-dimensional baselines do not rule out higher-order failure/task confounding.

TABLE S12. QUANTITATIVE ASSERTIONS EXCLUDED FOR MISSING BINDINGS.

Source	Missing binding for quantitative use
AgentPrint	Referenced supplementary mitigation result, procedure and population not located.
Weiss	No matched vendor-defense before/after outcome located; the artifact appendix reproduces an attack table.
Gu	No provider-change result with a matching comparison population located.
PromptEcho	MSS/threshold padding is described; exact threshold, population, cost denominator and defended outcome are unbound.

Sources: [30], [31], [32], [33]. No missing value is imputed; withholding these assertions does not remove the four families.

S6. Conclusion Sensitivity

Table S14 exposes direct-source dependence in the original snapshot. Within that snapshot, removing NetEcho withdraws the only downstream re-emission instance; removing AHB withdraws the measured mobile state-change instance, while WRA's privacy-oriented decoy result remains. Removing WRA withdraws that private-intent decoy result while AHB retains the task-state branch. UI Traces and SIGLEAK remain different surviving-feature instances, not replications. Removing Dangol withdraws its capture/live contrast; Early Bird, KVGov and SpliceDefense retain separate component examples. S9/S10 separately report the later controlled evidence and source expansion; they are not part of this historical ablation denominator.

Starting from the original 60/43/17 total/core/adjacent families, deleting AHB, WildAuth and coding fingerprints leaves 57/42/15; deleting its four diagnostics leaves 56/39/17; joint removal leaves 53/38/15. These are snapshot ablations, not the later 82-family collection. The snapshot's four diagnostic/semantic-boundary additions are

TABLE S13. SCOPED INTERVENTION DISPOSITIONS FOR THE 43 CORE FAMILIES IN THE ORIGINAL OCTOBER 8, 2026 SNAPSHOT. CATEGORIES ARE MUTUALLY EXCLUSIVE AT FAMILY LEVEL, NOT A COUNT OF SUCCESSFUL DEFENSES; LATER ADDITIONS ARE OUTSIDE THIS INVENTORY.

Source-scoped disposition	Families
Evaluated mitigation component/condition	16
Fixed common guardrail (not paired installation)	2
Other intervention / different security objective	4
No evaluation identified in inspected scope	13
Diagnostic goal: confidentiality defense inapplicable	4
Unbound assertions excluded from effect comparisons	4
Total	43

The four statements occur in their sources: AgentPrint supplementary analysis, Weiss vendor confirmation, Gu provider changes and PromptEcho padding figures. Quantitative result/run bindings remain unverified; all four families remain in the snapshot. Source-reported response, measured protection and missing measurement are distinct.

not premises of the cover/action contrast. For its five multifeature-trace studies, unresolved bindings bound D1 at three–five and D2 at two–five; removing AHB gives three–four and two–four. Early Bird calibration is a separate stage.

Excluding all four unbound defense quantities leaves the three deployment cases unchanged, because none uses those quantities as a premise. AHB's detector-run unknowns prevent an adaptive-detector claim but do not erase its separately reported task-success points. Its non-Fake online point is an additional Table 1 subcondition in the case ledger, not an extra independent experiment or an imputed value in the combined-condition inventory row. Whole-service costs, training access and missing utility admit no favorable imputation or ranking.

Narrowing H2 to matched tasks leaves task-independent identity unresolved; the document proxy does not authenticate capture. H3 supports only same-record offline adaptation. The original joint workflow endpoint remains negative, including the missing-score bounds and post-outcome sensitivity views. These positive, inconclusive, zero and negative outcomes are preserved with their original endpoint meanings.

S7. Retained Workflow Comparison

This finite model-assisted task compares a complete flat checklist (R), complete active structured checklist (T) and SCOPE receiver traversal (V). All receive the same evidence, ten checks and output schema in fresh `gpt-6-luna`/low sessions, with randomized request order within each root. The 512 primary roots use eight constructed templates, three queries per root, 11 nodes and four configurations. They are not independent natural-paper reviews or authenticated researcher participants.

The original joint endpoint requires the decision, exact configuration counts, evidence identifiers and schema all to pass. Correctness is 40.63% (R), 45.12–45.31% (T) and 32.16% (V). One T root is unscored; ranging its whole score over $[0, 1]$ gives V-minus-T of -13.15 to -12.96pp. Under fixed request-slot potential outcomes and no cross-request interference, the independent root contrasts lie in $[-1, 1]$. The two-sided finite-assignment Hoeffding radius is

$$h = \sqrt{2\log(2/0.05)/512} = 0.12004035.$$

The union over the missing score yields the conditional 95% envelope $[-25.16, -0.95]$pp. It is not a population-of-researchers interval; provider-state interference is unverified. All 1,590 scheduled slots, including 18 secondary source-file roots, were admitted; 1,589 are scorable, with the original overload HOLD and unknown usage retained.

Preserved negative result and analysis. V has 695 decision-right/count-wrong judgments versus T's 476, with no scorable schema/evidence-ID failure. Decision-only $+1.11$–$+1.30$pp changes the endpoint and cannot rescue the joint result. Six of eight templates have negative V-minus-T effects. Post-outcome flag-free and equality-pattern reductions remain negative (-12.43pp and -12.56 to -12.29pp); these are descriptive sensitivity views, not replacements. The 18 secondary source-file roots give -9.26pp descriptively. Exact-count semantics, repeated templates and fixed graph scale limit generalization.

The companion full protocol retains assignments, root/query records, missingness arithmetic, all decompositions and input checks. These data support no advantage for the tested workflow and no claim about human productivity or universal framework inferiority.

S8. Result-Pair and Six-Check Audit

This retrospective audit applies the paper's comparison rules to the existing 33 result records across 22 families. It adds explicit source judgments and deterministic aggregation, not new measurements or independent human coding. The machine-readable inventory retains every unordered pair ($\binom{33}{2} = 528$), the two proposed transfer directions, the coded reason and both source locators. Missing transfer evidence remains unresolved; known endpoint or mechanism conflicts are distinguished from missingness.

S8.1. Relevance and inference definitions

Six mechanism-question groups define 102 related pairs: traffic-envelope mitigation (66), cache-service mitigation (28), behavioral humanization (3), generator fingerprint variation (3), local-memory mitigation (1), and fixed guardrail/instruction variation (1). Access control, contention noise and filter-created observations remain singleton contextual groups. The other 426 pairs are outside these specified questions, rather than silently excluded as failures. These are retrospective relevance bins, not a sample of all possible comparisons or a literature-wide denominator.

A descriptive comparison is licensed when the target and receiver-facing outcome align while each metric, population and execution mode remains explicit. It licenses discussion, not numerical ranking. A paired effect additionally requires common defended test units and a verified fitting/matching

TABLE S14. LOAD-BEARING SOURCE ABLATION WITHIN THE ORIGINAL 60-FAMILY SNAPSHOT. ENTRIES IDENTIFY ITS ACTUAL DEPENDENCE; S9/S10 REPORT SUBSEQUENT CONTROLLED EVIDENCE AND SOURCE EXPANSION SEPARATELY.

Conclusion / source locator	What is supported	What is lost on source removal / NR dependence
Action preservation / UI Traces 6.1,Fig5	Recorded delays preserve structure and allow adaptation	Action-delay instance and one D1/D2 positive lost; exact release-point binding not needed for protocol claim.
Procedural preservation / SIGLEAK 4/C	Skill inference persists under common extraction defense	Skill-defense instance lost; no paired installation-effect claim.
Re-emission / NetEcho IV-C–E	Downstream reserialization can restore fine granularity	Only direct downstream re-emission instance lost; no multisource replication claimed.
State change / AHB 5.2.3/Table1/E.3	Fake actions can cause unintended UI transitions	Only executed mobile state-change instance lost; detector-run binding remains U.
Private-intent cover / WRA 8.1/Table8	Real/decoy execution with interleaved-domain inference evaluation lowers one-decoy E_{func}	Private-intent decoy instance lost; AHB state-effect instance remains. Same-condition terminal utility/cost are NR-R.
Mode split / Dangol §9.7, OHTTP paragraph after Table 5; 9.8–9.9/Table6	Capture frontier differs from live protocol and timing ablation	Concrete accuracy 0.17/0.30 pair and one D1/D2 positive lost.
Cost boundary / Early Bird V-A/B; Splice VII-C; Speculation 6.1; KVGov 5.9	Separate component and deployment costs	Removing one withdraws its example, not other boundary instances; full service costs remain unknown.
Generator dependence / UI Traces, FP-Agent, traffic benchmark, LeakyLMs	Task, browser, topology and calibration tests differ	Each source's axis withdrawn when removed; no pooled portability estimate.

TABLE S15. DESCRIPTIVE COMPARISON WITHIN DECLARED SUBSETS. PERCENTAGES HAVE THE DISPLAYED ROW DENOMINATOR, NOT ALL 528 PAIRS.

Subset	Compatible	Incompatible	Unresolved
Same family ($n = 13$)	7 (53.8%)	6 (46.2%)	0
Related ($n = 102$)	16 (15.7%)	80 (78.4%)	6 (5.9%)

relation. Directional transfer asks whether a source operating point can be reused unchanged at the destination condition. An absent bridge is unresolved; a demonstrated wrong object or conflicting operating point blocks that particular reuse. Pairs can therefore receive different outcomes for the three inferences.

The 13 same-family pairs provide the least arbitrary comparison subset (Table S15). Seven compare aligned outcomes and six substitute different primitives or cost objects. Of the seven aligned-outcome pairs, four have known differences in collection/execution populations and three lack a verified paired-test/fitting binding. None licenses subtraction across those two records. This is separate from the positive paired adaptation contrasts already recorded *inside* UI-delay-offline and DANGOL-capture. For unchanged deployment transfer, five same-family pairs have a positively conflicting requested reuse (10 directions) and eight lack a bridge (16 directions). Across the 102 related pairs, the 204 directions comprise 10 blocked and 194 unresolved cases. No cross-record positive bridge is established by this inventory; source-local permissions remain usable.

For example, transformed Dangol captures and its live loopback stack report different operating points (approximately 0.17/16.2× bytes versus 0.30/13× bytes/11× latency). They can be described together as workflow-recovery evidence but not as a paired deployment improvement. SpliceDefense's CTBF signal suppression and QCP-only

TABLE S16. SIX CHECKS OVER 16 EVALUATED-MITIGATION FAMILIES. COUNTS APPLY TO THE SOURCE-SCOPED CLAIM; NO COMPOSITE QUALITY SCORE IS COMPUTED.

Check	Bound	Partial	NR-V	NA
1. Target/protection relation	16	0	0	0
2. Receiver fields/permissions	13	3	0	0
3. Placement/action dispatch	13	2	1	0
4. Protected fit/test/access	0	4	5	7
5. Tested workloads/shifts	14	2	0	0
6. Joint inference/utility/cost	0	16	0	0

Bound: the required item is source-bound for the scoped claim. Partial: supported elements coexist with an unbound element. Check4 includes the observer's training-data access and matching, not just existence of a fitted model. Check6 asks for an end-to-end deployment join, not a requirement that every component study perform one. No whole-check cell is solely NR-R; individual unreported subfields remain NR-R in the 96-cell inventory.

cost likewise answer different component questions. The counts expose the consequences of the declared comparison, not an observed rate of community misreading or a benchmark of the field schema.

S8.2. Six checks and source corrections

Exactly the 16 families with an evaluated-mitigation disposition contribute six cells each. Table S16 summarizes these 96 judgments; the CSV retains the result IDs, source scopes, reasons and separate utility/cost objects. A partial item preserves both supported and missing elements, rather than treating a useful component measurement as no evidence.

The joint deployment item is partial for all 16 because the recorded component, replay or separately collected results do not jointly bind every required deployment object. This does not impose an agent-level experiment on stud-

ies of individual components. AHB binds detectability and objective task success but not execution cost; Dangol's live stack binds inference and protocol costs but not objective task utility; Speculation reports a padding score and token-payload ratio. In the adaptation item, partial status also reflects unbound observer access or fitted-test details even when source-local protected fitting is established.

The audit corrects four historical interpretations through an additive overlay. Speculation uses a Random Forest, and Time Will Tell fits input-length-conditioned thresholds; their defended-fit/run bindings are NR-V, not inapplicable merely because the attack was described as matching or a count rule [34], [35]. SafeKV Figure 12 explicitly uses a five-second post-trigger window, resolving that field. SafeKV also reports separate serving-performance experiments: in the stated multi-turn comparison against full isolation, Qwen's extra TTFT falls from 50.41% to 11.74%, and DeepSeek's from 118% to 34.28%; at 16 requests/s its tested configurations achieve 1.36–2.66× the token throughput of Cache-Partition [36]. These are service measurements, not only detector latency. They retain their own workload/condition rather than becoming a joint leakage–cost point with Figure 12. Original frozen records remain available alongside the corrected overlay.

S8.3. Reproduction and interpretation

The package's `decision-audit/` directory contains the rules, explicit semantic coding, all-pair inventory, 1,056 directional rows, 96 checklist cells, source corrections and read-only checker. The checker verifies unique coverage, input hashes, source/result ownership and reproducibility of the saved counts without fitting or network access. Original full texts are not redistributed; URLs, versions, locators and hashes permit source lookup. Counts are descriptive: no significance test, pooled efficacy score, causal effect, reviewer-error rate or independent human agreement is estimated. Thus the audit quantifies what the source bindings permit while keeping the underlying evidence and comparison question visible.

S9. Controlled Intervention Details

The frozen controlled study uses `gpt-6-luna`, low reasoning, through the OpenAI Codex subscription interface. Each response is obtained from an ephemeral thread with no execution environment, available tools, workspace roots or inherited project instructions. Only synthetic public task inputs are transmitted. Original final answers, request/turn/usage records, model attestations and process-closure receipts accompany the results. Released transport projections omit account events and local identifiers, with hashes linking them to retained originals. There are 72 application admissions, 72 completed responses, zero application retries and 195,927 observed tokens. Collection takes 472.6 seconds. Provider-internal retries and underlying network-request counts are not exposed by this interface.

Receiver experiment. The fixed seed 20261021 assigns 24 topic pairs to twelve training and twelve test topics; labels are brief and expanded private instructions. Each saved text traverses each arm once in a shuffled schedule, producing 144 trials and 288 observer/site views. TLS1.2 with ECDHE-RSA-AES128-GCM-SHA256 permits explicit parsing of application record types. The relays record encrypted byte streams, direction and local read timestamps; they cannot read the explanation. Four-byte application length headers are inside encryption. Padded frames have 1,024 application bytes and 24 bytes of GCM nonce/tag overhead, plus the five-byte TLS record header. Sixteen such records therefore cost 16,848 bytes. No padding-budget overflow or content-integrity failure occurs.

The attacker fits one length threshold and direction on training topics, including constant-prediction candidates with fixed tie-breaking. It is retrained independently for each protected arm/site. Held-out classifications are 24/24 at unpadded links and 12/24 at padded links. Upstream-only padding leaves terminal predictions unchanged; final-link padding changes terminal accuracy by -50 percentage points. Every paired test topic has the same difference, making the saved topic-bootstrap intervals degenerate. These intervals reflect this fixed workload and estimator; they do not establish a population-wide error bound. The definitive mechanism check is that the captured padded application-record feature is constant across all 48 outputs.

Median end-to-end replay time is 25.97ms without padding, 27.33ms with upstream-only padding and 28.52ms with both links padded. These descriptive local measurements include two TLS handshakes and frontend work, but exclude model generation and remote deployment latency. The wire multiplier is 808,704/34,606, approximately 23.37 per padded link across the 48 outputs. All 144 terminal content hashes match the source, and all owned socket threads join.

Execution experiment. The 24 tasks split evenly between inventory and calendar domains. Twelve cover writes overlap the target resource; six use a disjoint protected resource and six use a disjoint spare resource. Three of the six overlapping inventory tasks have spare stock, while three have exactly the requested stock. All six overlapping calendar writes occupy the requested slot before the real booking. Cover writes have valid actor/confirmation parameters; failures therefore concern state semantics rather than malformed requests. All 24 model plans are valid and contain one write. No plan is dropped or repaired.

Across the five routes, 120 task/arm rows are retained as paired observations of 24 tasks. Live-write successes are 6/12 inventory and 3/12 calendar. Overlapping inventory succeeds in 3/6; overlapping calendar in 0/6. Each domain's disjoint cover succeeds on all six requested goals, but three protected-resource violations reduce full success to 3/6. This separates objective completion, hard constraints and exact state preservation. All four cover implementations retain their actual local tool call counts and elapsed time; local microseconds are not remote API costs.

The live-write full-success estimate is 37.5% (Wilson

95% interval 21.2–57.3%), compared with 100% (86.2–100%) for the other routes. The prespecified paired statistic resamples whole tasks 10,000 times with seed 20261021 and gives a difference of -62.5 points, interval $[-79.2, -41.7]$. These intervals describe variation among constructed examples rather than a natural task population. The deterministic state-equality results apply to the implemented pure-read and whole-clone-isolation routes. Neither state equality nor utility implies indistinguishability to a privacy attacker.

Verification and retained outcomes. Eighteen no-model tests cover protocol failures, missing usage, tool authorization and confirmation, malformed plans, state isolation, TLS parsing, content preservation and cleanup. The separate post-collection audit reads raw model events, ciphertext captures and before/after states and checks saved numerical results. It does not request new model responses or replace observations. Original empty/failed retail runs and the negative workflow-audit endpoint remain in Supplements S3/S7. New controlled results support the placement decisions for their explicit system and task constructions.

S10. Completed Finite Retrieval Update

This update fixes a feasible retrieval endpoint and completes its dispositions rather than treating the historical broad-hit counts as a screened population. Four Semantic Scholar bulk queries and two native arXiv searches were completed on October 21, 2026. Their 615 occurrences deduplicate to 565 records. The merged family index accepts 15 new core and seven selected adjacent families, giving 82 families: 58 core (20 workflow-origin, 38 inherited) and 24 adjacent. Thirteen new core versions predate the original October 8 endpoint; Physical Edge (October 9) and PAGE404 (October 14) explicitly extend it. The original 60-family index and 33-result/66-field inventory remain separate snapshots.

S10.1. Executable queries and accounting

For Semantic Scholar, the actual bulk-search expressions below use + for conjunction, | for alternatives and quotation marks for phrases. The publication window is 2023–2026; returned metadata and abstracts, query parameters, pagination/completion receipts and exact source locators are saved in the query log. These searches use the service's title/abstract retrieval behavior, not publisher full-text search.

S1R (116): (agent | workflow) + (LLM | "language model") + ("side channel" | "side-channel" | fingerprinting).
S2 (120): (LLM | "language model") + (traffic | streaming) + (leakage | side-channel | fingerprinting).
S3 (15): (LLM | "language model") + (padding | "re-emission") + (privacy | leakage | side-channel).
S4 (287): (agent | LLM) + (decoy | "cover traffic" | "fake actions" | "cover actions").

Native arXiv advanced search used all fields, Computer Science including cross-lists, original-submission dates January 1, 2023–October 21, 2026, abstracts shown, and a 200-result page limit. The two conjunctions were `"side-channel" AND "large language"` (50) and `"side-channel" AND agent` (27). Each returned set fit on its saved page. The Semantic Scholar queries yield 538 occurrences/522 unique IDs; arXiv adds 43 further unique records, producing 565 overall. Exact native URLs and full result metadata are retained.

TABLE S17. DISPOSITION FLOW FOR THE COMPLETED 565-RECORD FRAME. SOURCE-SCOPED EXAMINATION IS NOT SYNONYMOUS WITH NEW CORE INCLUSION.

Stage or final route	Count
Complete-query occurrences	615
Cross-query duplicates merged	50
Unique records with an explicit route	565
Excluded: no hidden auxiliary target	217
Excluded: outside LLM execution	156
Excluded: different inference target	19
Existing core / adjacent families recovered	24 / 1
New source-bound core families	15
Adjacent source-reviewed / metadata-only contexts	30 / 28
Other security-objective background	53
Source-reviewed background / proposal metadata	4 / 1
Missing abstract / unavailable primary full text	16 / 1

Forty-nine candidate rows received source-scoped examination: 15 new core, 30 adjacent contexts and four background sources. Seven of the 30 adjacent contexts are selected for the merged index; the other contexts remain reasoned routes, not additional admitted families. Sixteen missing abstracts and the inaccessible AgentCard primary full text remain unresolved material requests. Perplexity Search supplied additional leads and counterchecks, but its eight saved dispositions are kept outside the 565-record query denominator; primary-source claims are verified against their own records.

The broader Semantic Scholar S1 acquired 999 of 2,104 reported hits before refinement; it is not a complete screened frame. Its complete refinement S1R is a subsequent, documented scope decision. S4R returned HTTP429, arXiv API attempts failed, and DBLP access returned errors/verification rather than a usable export. A third native arXiv query timed out without a verified set; a fourth was not submitted. None is recorded as a zero-result search. These access outcomes and the historical IEEE/ACM acquisition gaps remain in the log.

S10.2. Boundary checks and extraction

The fixed 24-known-paper set has 17 query recoveries (nine Semantic Scholar, fourteen arXiv, with overlap); the original 60-family set has 25 (fourteen and nineteen respectively). These checks show the additional route's concrete contribution. They do not establish calibrated database recall: index eligibility for each non-return has not been authenticated.

The exclusion check uses seed 20261021: all 177 exclusions matching the recorded high-risk title/abstract pattern, plus 27 other reason-stratified samples, capped at ten per

reason. Four of 204 rows were rerouted. MIRAGE becomes one new diagnostic core; InputSnatch resolves to an already included title/version alias; two become adjacent contexts. The final disposition table incorporates these corrections. Because the check is enriched and uses the same AI-assisted workflow, four of 204 is an observed correction count, not a population miss-rate or independent-human agreement estimate.

Every new core row records observer, signal, target/protection relation, evaluation conditions, intervention, utility, cost, RQ contribution, version, locator, hash and NR status. Lidar's shared action roles and checked outcomes exclude raw command strings, paths and answers; feature importance is not treated as an isolated attack. SplitDecoy qualifies specifically through zero/nonzero returned-gradient support. Direct visual-token and gradient-value reconstruction remain adjacent under the auxiliary-only criterion. MIRAGE's privileged diagnostic is not recast as a low-permission privacy attack.

S10.3. What changes in the synthesis?

For RQ1–2, the update adds tool-policy audits, structural campaign linkage, routing histories, tokenizer counts and distinct physical/sparsity/page-migration observers. These are different targets and permissions, not repeated demonstrations of each case. For RQ3, SplitDecoy supplies another outgoing path whose protection must be checked, while Kotevska et al.'s six-element claim register is close prior framework work; a tuple or generic composition warning is therefore not claimed as our novelty.

For RQ4, the positive defense evidence becomes stronger and more specific. CachePrune evaluates sensitive-token protection, QA/summarization quality and serving performance under selective sharing. Contextual reconstruction and detector false negatives retain their own endpoints. OTRO's optional epoch padding reduces additional access-count information from 3.94 to 0.04 bits conditional on byte length in a 20K-prompt model. Its TTFT text reports 4.59% mean overhead for Llama-3.1, whereas the abstract rounds this to at most 4.5%; the main TTFT figures explicitly omit padding. The padding cost is a separate calculation (up to about 2.8ms per epoch), not a measured end-to-end padded TTFT frontier. These details permit useful local claims without erasing residual byte-length information.

Jiao's packet-only and fused interrupt/packet endpoints have different permissions; SparLeak's separately measured attacker-monitoring slowdown is not the cost of its sparse-token defense. Shape and Substance's padding estimate is not a deployed paired intervention. These extraction differences change which cost can enter a deployment comparison. Consequently, the original 33-result inventory cannot support a collection-wide absence claim. Its comparisons remain reproducible for that subset; the added families supply new counterconditions and positive cases. The new controlled experiments separately test receiving and execution placement, without replacing old failed or negative observations.

The machine-readable merged index preserves every field of the original 60 rows and appends the 22 additions. The query log, per-record reasons and short evidence fragments, source hashes, correction ledger and sensitivity notes accompany it. Full third-party sources are identified by version/URL, not redistributed in the anonymous package.

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